

Omri Ben-Dov

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Professional Summary

PhD candidate at the Max Planck Institute for Intelligent Systems (expected Aug 2026) working on principled objectives and optimization for generative modeling and learning under strategic behavior. I build divergence- and likelihood-based training methods that improve convergence and stability for density estimation and simulation-based inference (WACV 2024; preprint). I also develop collective-action approaches to improve fairness with minimal accuracy cost under coordinated user behavior (ICML 2024; NeurIPS 2025 workshop oral).

Research Experience

Max Planck Institute for Intelligent Systems

Tübingen, Germany

PhD student

2021–Present, expected August 2026

- **Density estimation:** Developed a symmetric divergence minimization objective; evaluation on UCI datasets and synthetic benchmarks, better NLL and no overfitting versus MLE variants in few-data regime and more stable than GANs.
- **Collective action in ML (ICML 2024; NeurIPS 2025 workshop oral):** Developed collective-action interventions that improve fairness with minimal accuracy cost versus post-processing methods across standard benchmarks (COMPAS, Adult, HSLS, Waterbirds, CivilComments, ACS Income).
- **Research engineering:** Built a Hydra-based experimentation pipeline and ran ablations/sweeps on an HTCCondor cluster.
- **Articulated 3D modeling:** Learned an articulated 3D model of a laboratory rats from MRI imaging (CVPR workshop).

Weizmann Institute of Science

Rehovot, Israel

Research Assistant

2021

- Applied signal processing and matched filtering on gravitational waves data to detect binary black holes.

The Hebrew University of Jerusalem

Jerusalem, Israel

MSc in Micro-Flight Group

2018–2021

- Developed an algorithm to estimate kinematic parameters of free-flying flies from video using a 3D model.

Selected Publications and Preprints

- **Omri Ben-Dov**, Luiz F. O. Chamon (2025). **Adaptive Symmetrization of the KL Divergence**. arXiv: 2511.11159.
- **Omri Ben-Dov**, Samira Samadi, Amartya Sanyal, Alexandru Tifrea (2025). **Fairness for the People, by the People: Minority Collective Action**. *Oral Presentation in the Algorithmic Collective Action workshop, NeurIPS 2025*. arXiv: 2508.15374.
- **Omri Ben-Dov**, Jake Fawkes, Samira Samadi, Amartya Sanyal (2024). **The Role of Learning Algorithms in Collective Action**. In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. PMLR, pp. 3443–3461.
- **Omri Ben-Dov**, Pravir Singh Gupta, Victoria Abrevaya, Michael J. Black, Partha Ghosh (2024). **Adversarial Likelihood Estimation With One-Way Flows**. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3779–3788.

Education

Max Planck Institute for Intelligent Systems

Tübingen, Germany

PhD in Computer Science, Expected Graduation: August 2026

2021–Present

The Hebrew University of Jerusalem

Jerusalem, Israel

MSc in Computer Science

2018–2021

Thesis: "Automatic Pose Estimation of Free-Flying Fruit Flies".

Magna cum laude.

Technical Skills

Languages: Python, C/C++, MATLAB

ML/Stats: PyTorch, Scikit-learn

Experimentation: Hydra

Compute: HTCCondor (cluster experimentation)

Tools: Git, Jupyter, L^AT_EX

Leadership and Service

PhD Student Representative, Gender Equality Program, MPI-IS (2022–2024): Collaborated with institute leadership to implement gender equality and diversity initiatives.

Committee Member, Athena Group, MPI-IS (2023–2025): Organizing monthly lunches at the Tübingen site to discuss and promote diversity and inclusion.

Awards and Honors

Silver Reviewer: ICML 2026

Top Reviewer: NeurIPS 2025

Dean's Honor List, The Hebrew University of Jerusalem (MSc): 2021

Weizmann Institute *Amos de-Shalit Summer School* for excellent students in Physics: 2017

Excellent Students in Computer Science Program (BSc): 2016–2018